

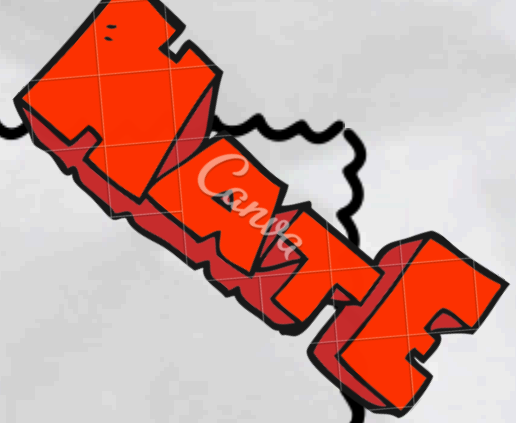
FORENSIC RESEARCH PROJECT

METHODOLOGICAL FRAMEWORK FOR FINDING HATEFUL CONTENT

FINAL PRESENTATION
FT. SCHAPPELLE CORBY

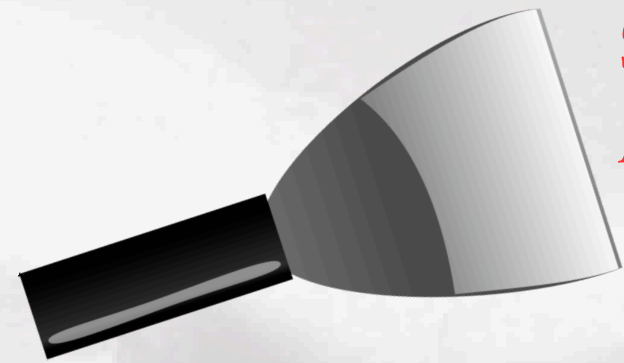
BACKGROUND

- GLOBAL CASE, DEBATES ON JUSTICE & MEDIA INFLUENCE
- REDDIT & X DISCUSSIONS ANALYSED
- USER OPINIONS, NARRATIVES & MEDIA REACTIONS EXAMINED
- SOCIAL MEDIA SHAPES CRIME & JUSTICE DISCOURSE

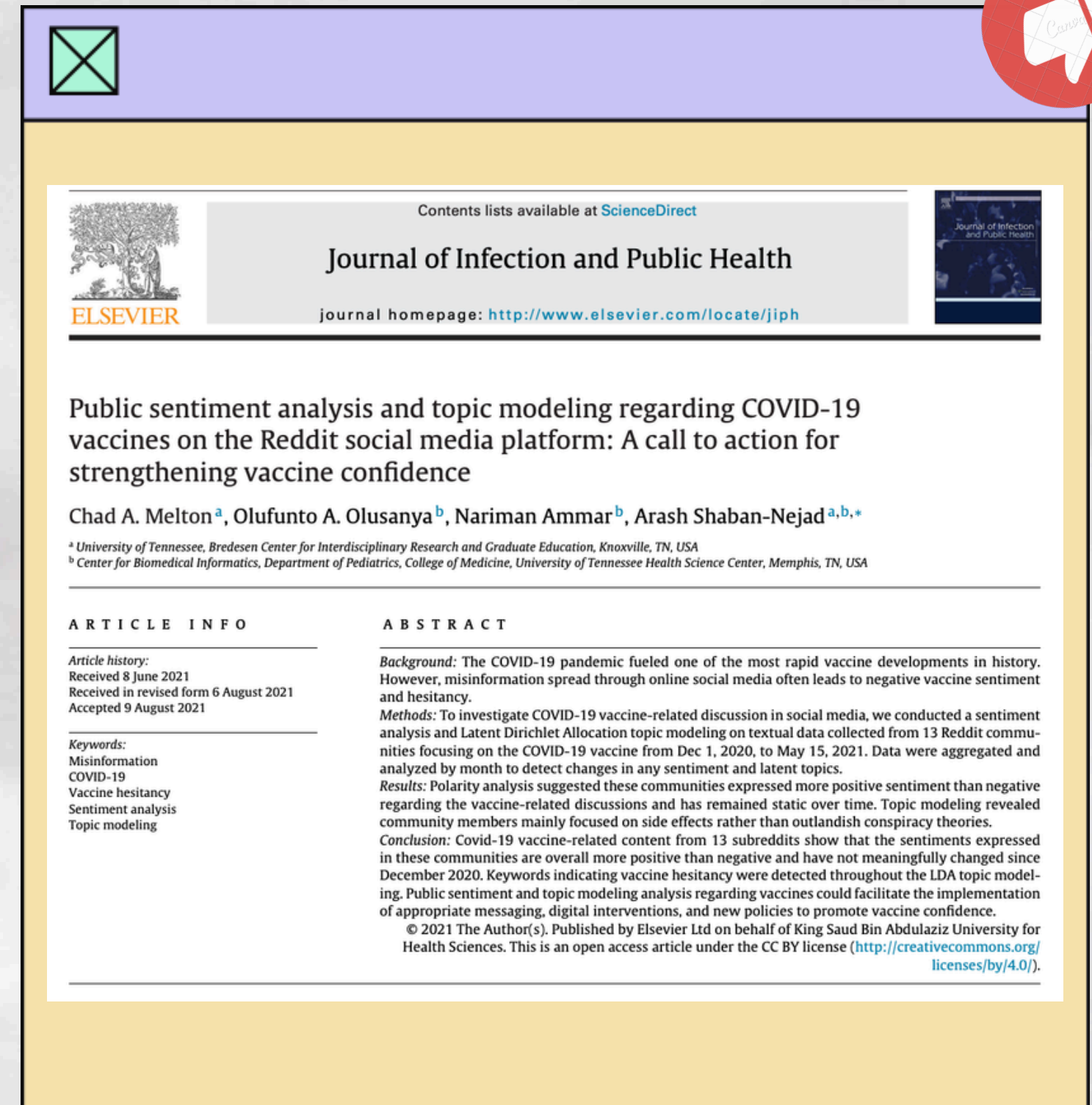
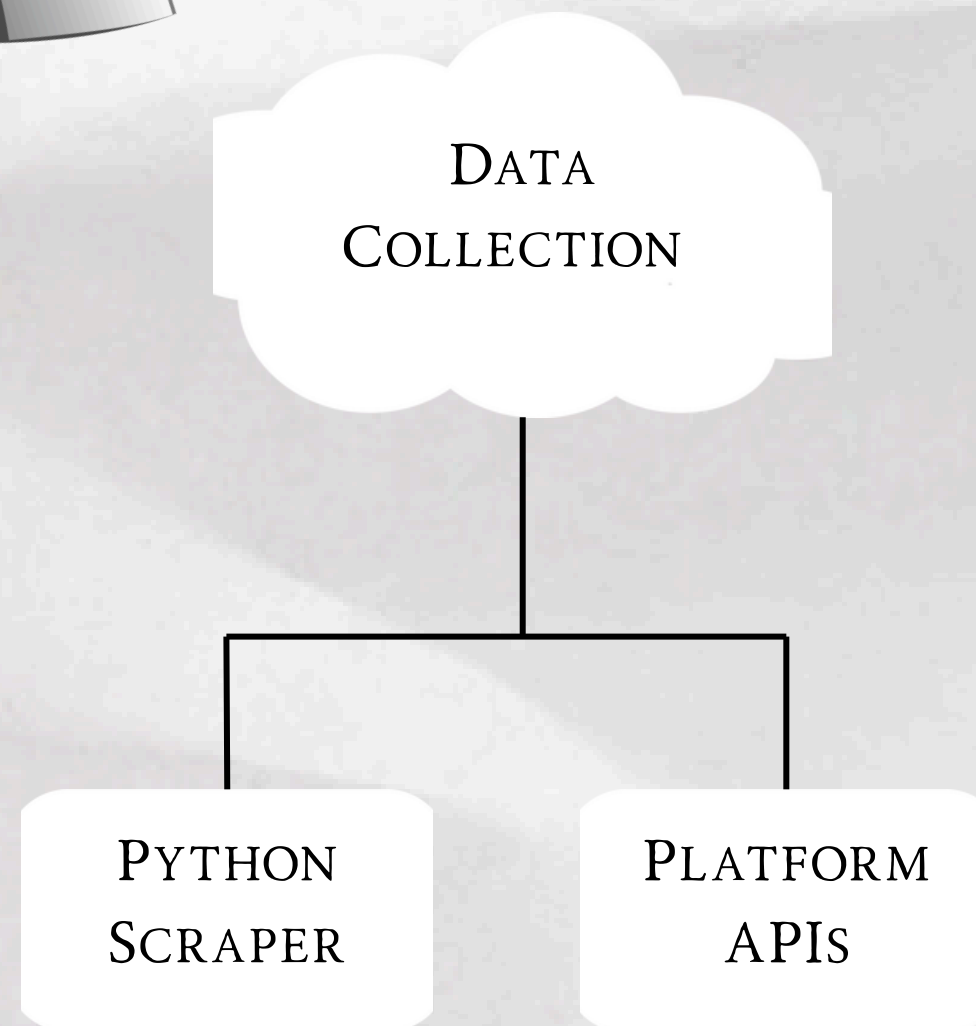


WEB SCRAPING

SOCIAL MEDIA: KEY SPACE FOR NEWS, DISCUSSION, AND
OPINION IN AUSTRALIA (ACMA, 2024)



SCRAPERS EXTRACT PUBLICLY
AVAILABLE WEBSITE CONTENT



[1] MELTON, C. A., & OLUSANYA, O. A. (2021, AUGUST 9). PUBLIC SENTIMENT ANALYSIS AND TOPIC MODELING REGARDING COVID-19 VACCINES ON THE REDDIT SOCIAL MEDIA PLATFORM: A CALL TO ACTION FOR STRENGTHENING VACCINE CONFIDENCE. JOURNAL OF INFECTION AND PUBLIC HEALTH. RETRIEVED MAY 13, 2026, FROM [HTTPS://ARXIV.ORG/PDF/2108.13293](https://arxiv.org/pdf/2108.13293)

METHODS

EXPERIMENTAL METHOD

- PLATFORM SELECTION
- REGULATION REVIEW
- DATA SCRAPING (POSTS/COMMENTS)
- DATA STORED IN JSON FORMAT

DATA PROCESSING & ANALYSIS

- TEXT CLEANING & NORMALISATION
- TOKENISATION
- SCRAPING METHODS COMPARED
- EVALUATED USING METRICS

AIM	OBJECTIVES
TO DEVELOP A FRAMEWORK TO FIND SPECIFIC INFORMATION RELATING TO A CASE AND STORE AND LABEL IT ETHICALLY	IDENTIFY CLEAR WEB PLATFORMS HOSTING PUBLICLY ACCESSIBLE COMMUNITY FORUMS
	EVALUATE TOOLS AND METHODS FOR THE ACQUISITION OF ONLINE TEXT DATA UNDER PLATFORM AND ETHICAL CONSTRAINTS





HATE SPEECH CATEGORIES

×

DEFINITION

REFERS TO OFFENSIVE DISCOURSE
TARGETING A GROUP OR AN
INDIVIDUAL BASED ON INHERENT
CHARACTERISTICS AND THAT MAY
THREATEN SOCIAL PEACE

– □ ×

CASE RELATED

⋮
NAMES & LOCATION

– □ ×

LEGALITY-BASED

⋮
ARGUE GUILT

– □ ×

CRIME-BASED

⋮
CRIME
INVOLVEMENT

– □ ×

CLASS-BASED

⋮
SOCIAL
JUDGEMENT

– □ ×

FAMILIAL-BASED

⋮
TARGETING
FAMILY

– □ ×

AUTHORITY-
BASED

⋮
TARGETING
SYSTEM

– □ ×

MORAL
CONDEMNATION

⋮
JUDGING ACTIONS

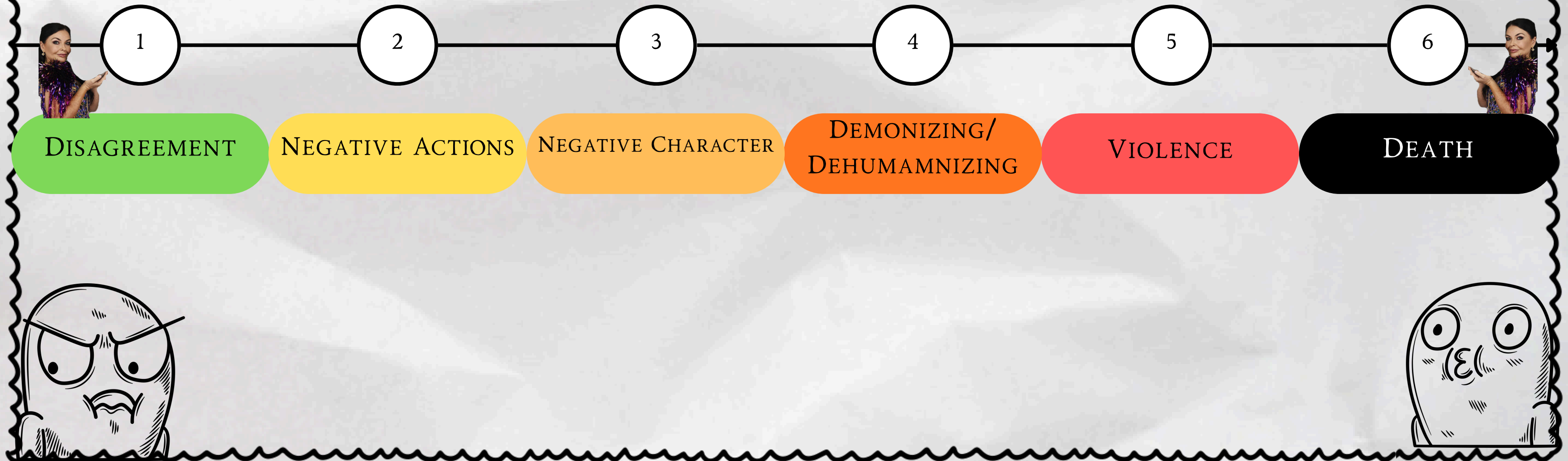
– □ ×

INTELLIGENCE-
BASED

⋮
DIRECT
INSULT

MEASURING HATE SPEECH

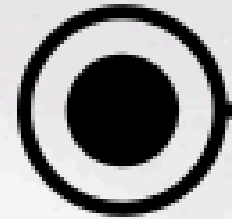
- TO CLASSIFY SEVERITY OF HATE SPEECH FROM SCRAPED POSTS AND COMMENTS
- TO IDENTIFY USEFUL INSIGHTS BETWEEN FREQUENCY AND SEVERITY



METHODOLOGY - PYTHON SCRAPERS



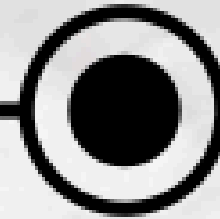
SCRAPY SPIDER SCRIPT BUILT TO SCRAPE
SCHAPPELLE CORBY-RELATED COMMENTS FROM
WEBSITES.



16/03/2026



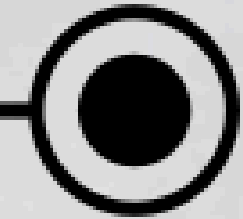
OBTAINED RESULTS FROM THE
PYTHON SCRAPER INTO JSON
FORMAT



20/03/2026



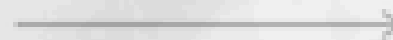
ANALYSED RESULTS IN
RELATION TO HATE CLASS
TERMS BEING USED



31/03/2026

What it is

- Automated data extraction via Python script
- Extracts structured data directly from websites



How it works

- 1 Sends requests to web pages
- 2 Parses HTML / dynamic content
- 3 Extracts relevant data
- 4 Outputs data in JSON format

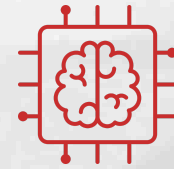
Output ready for analysis



METHODOLOGY - WEB SCRAPERS



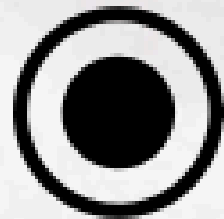
RESEARCH AND IDENTIFY WEB SCRAPERS TO BE POTENTIALLY USED



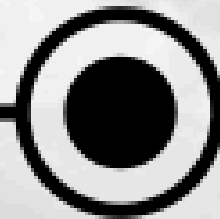
RECEIVED BURNER ACCOUNT DETAILS TO ACCESS WEB SCRAPING PLATFORMS AND CONDUCT SCRAPING



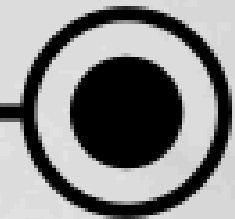
COMPILED DATA FROM SCRAPERS AND MADE COMMENTS READY FOR CLEANING AND ANALYSIS



3/02/2026



20-31/03/2026



14/04/2026

What it is

- Browser-based, visual interface
- No coding required

Tools used

BrowseAI

Octoparse

ParseHub

Apify

How it works

- 1 Selects elements from web pages
- 2 Automatically extracts data
- 3 Exports results in JSON format

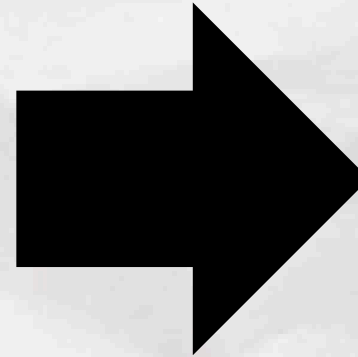
Output ready for analysis



METHODOLOGY - DATA CLEANING

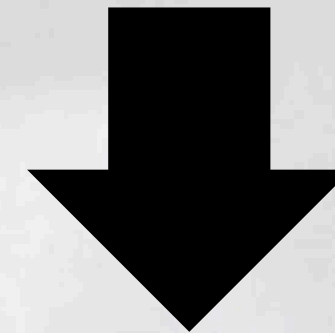
RAW JSON DATA

- SCRAPED WEB DATA
- NESTED STRUCTURE



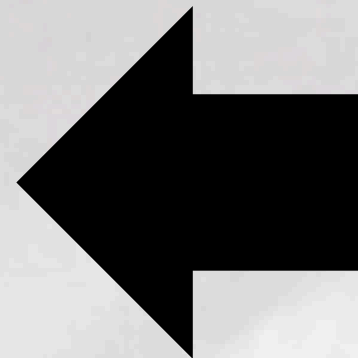
PYTHON SCRIPT

- AUTOMATED CLEANING
- DATA FORMATTING



FINAL DATASETS

- STRUCTURED
- ANALYSIS-READY



CLEANING TASKS

- REMOVING EMOJIS
- ALL LOWERCASE TEXT
- REMOVE SPECIAL CHARACTERS



PYTHON SCRAPER - DATA COLLECTION

●● SITES SCRAPPED

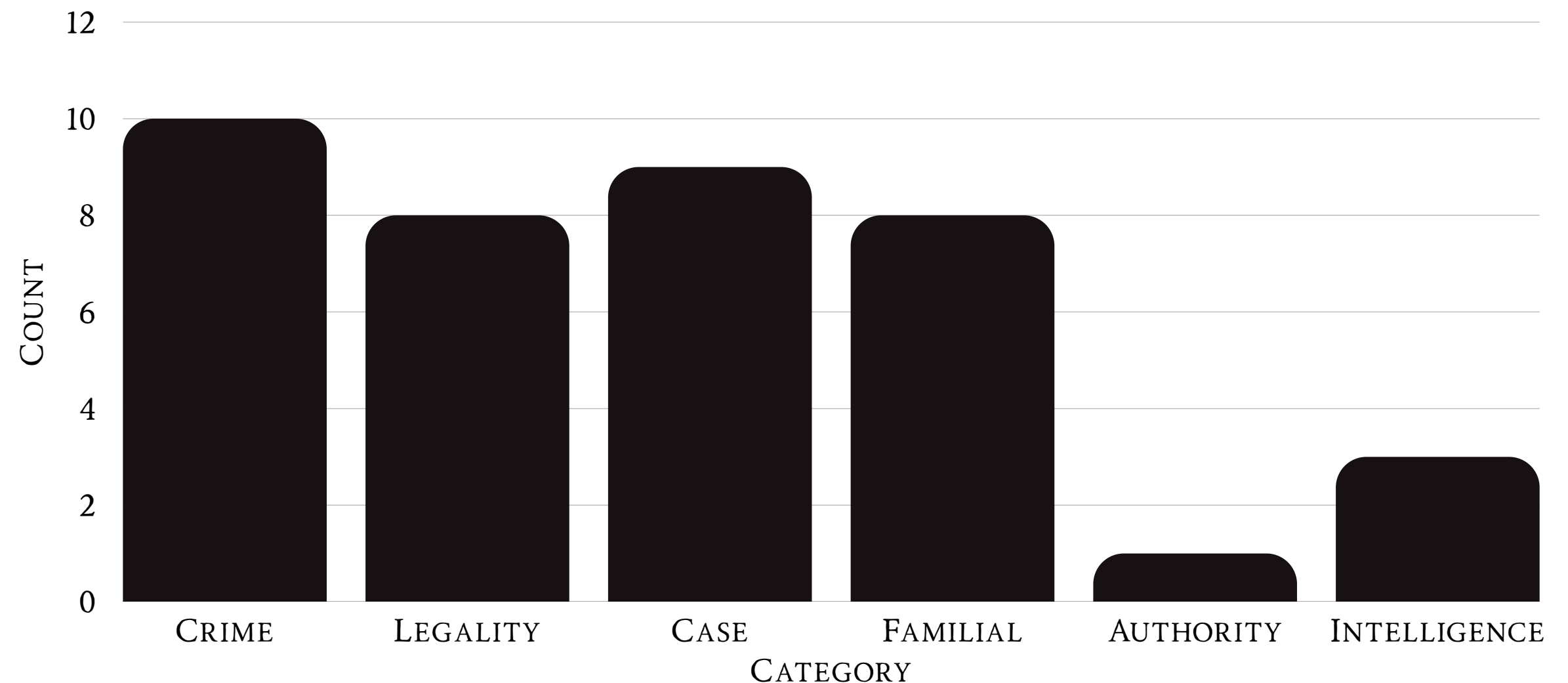
◆ 9 NEWS

◆ ABC

◆ PERTH NOW

◆ WIN NEWS

BAR CHART OF CATEGORIES COLLECTED FROM PYTHON
"SCRAPY" WEB SCRAPER

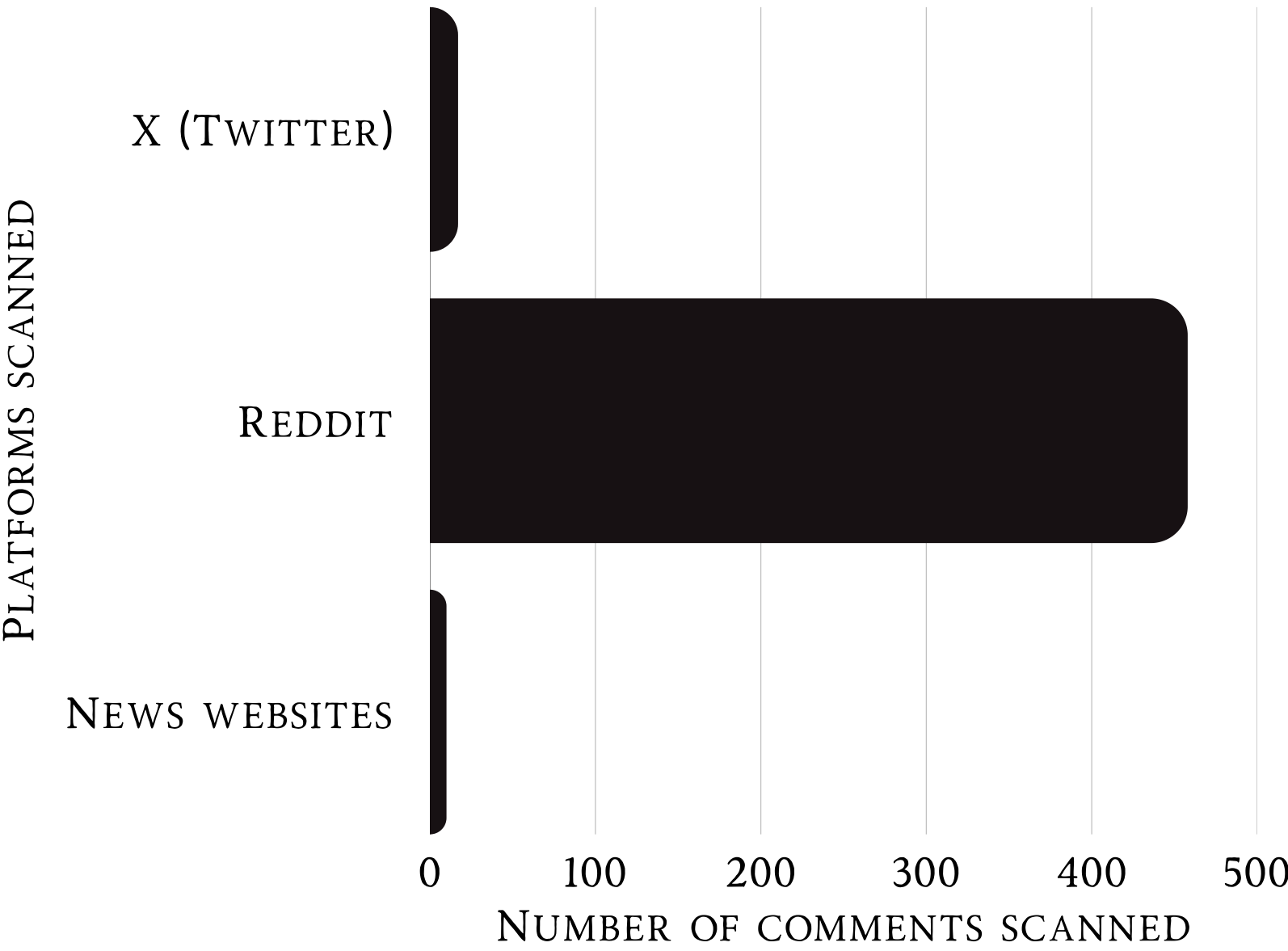


FACTUAL CATEGORIES DOMINATE DUE TO NEWS-ONLY SCRAPING

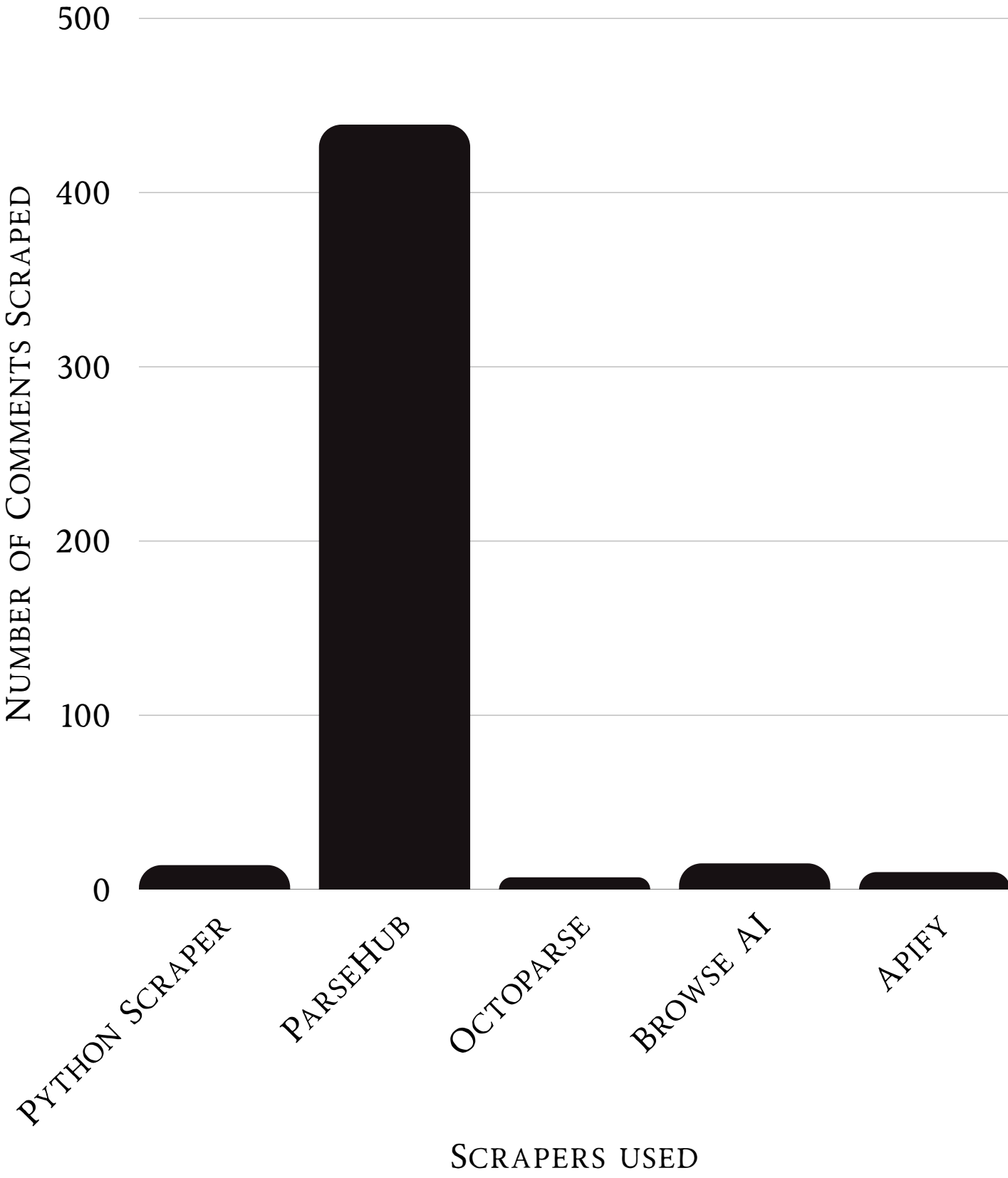


RESULTS AND ANALYSIS

NUMBER OF SCANNED
COMMENTS ACROSS SITES

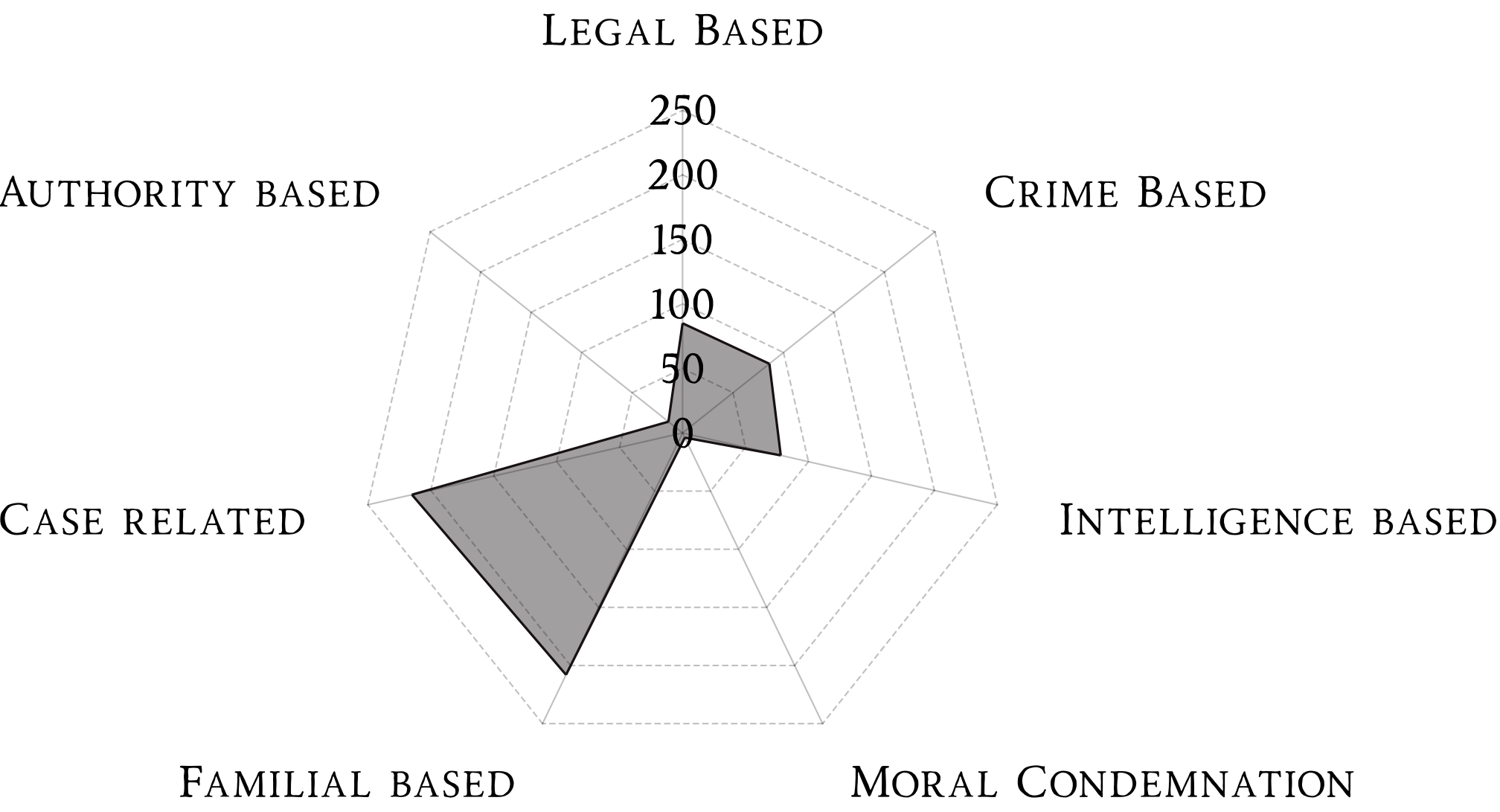


NUMBER OF COMMENTS PER
SCRAPER



RESULTS AND ANALYSIS

NUMBER OF COMMENTS
FOUND IN EACH
CATEGORY



Correlation of Types of Words Heatmap

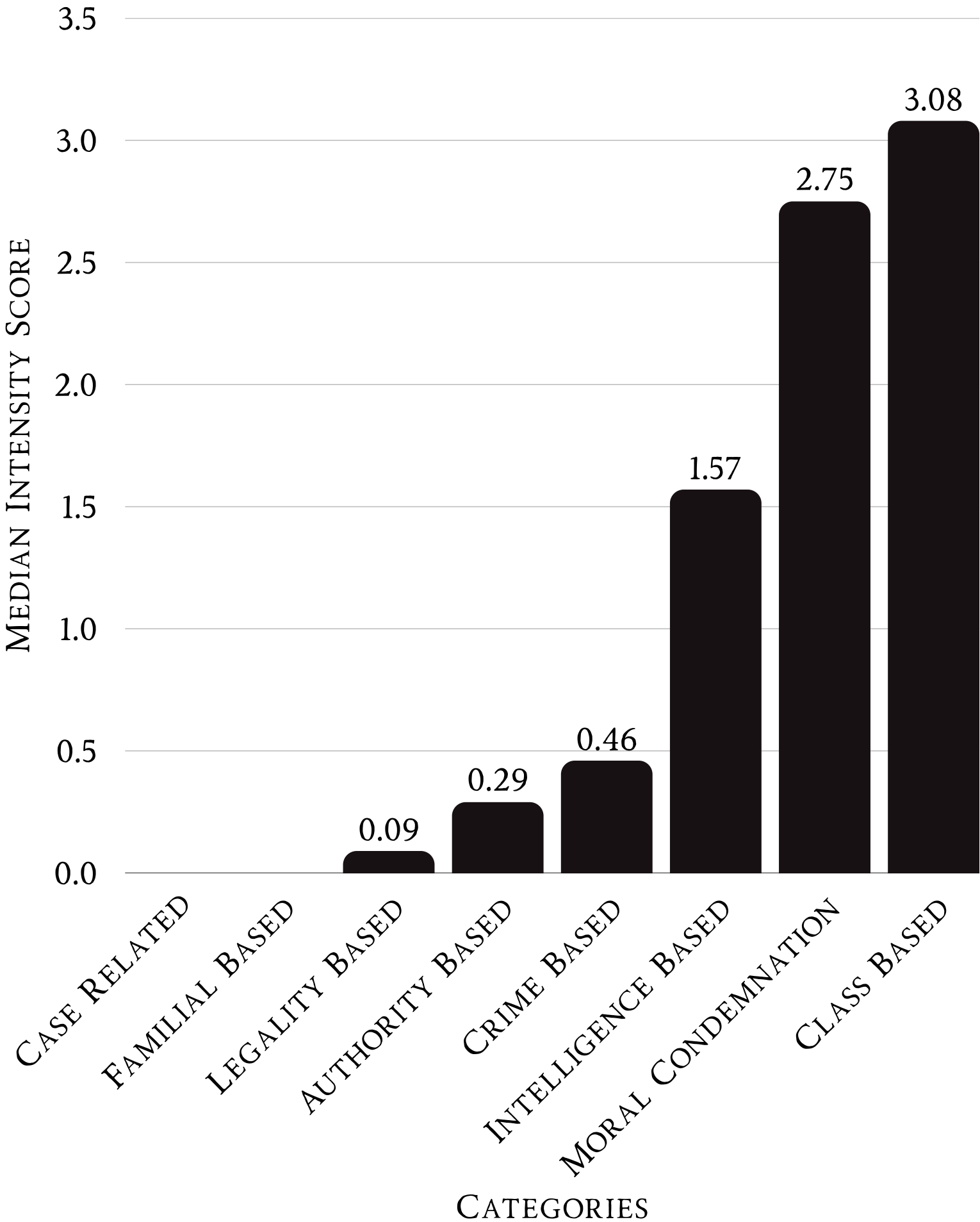
frequency of themes found together (% of rows where both appear)

	Authority	Case	Class	Crime	Familial	Intelligence	Legality	Moral
Authority	100	93	14	50	43	86	57	50
Case	93	100	62	40	97	45	40	75
Class	14	62	100	28	34	34	34	25
Crime	50	40	28	100	27	49	68	75
Familial	43	97	34	27	100	38	35	50
Intelligence	86	45	34	49	38	100	68	75
Legality	57	40	34	68	35	68	100	75
Moral	50	75	25	75	50	75	75	100

HATE SCALE STATS

CATEGORIES	COUNT OF TERMS	PERCENTAGE OF GRAND TOTAL	POPULAR TERM
MORAL CONDEMNATION	4	0.44%	DISGUSTING
AUTHORITY BASED	14	1.53%	AFP
CLASS BASED	36	3.93%	BOGAN
INTELLIGENCE BASED	81	8.83%	AF
CRIME BASED	111	12.10%	DRUGS
LEGALITY BASED	112	12.21%	GUILTY
FAMILIAL BASED	231	25.19%	CORBY
CASE RELATED	328	35.77%	SCHAPELLE CORBY

RELATIVE MEDIAN HATE INTENSITY BY CATEGORY

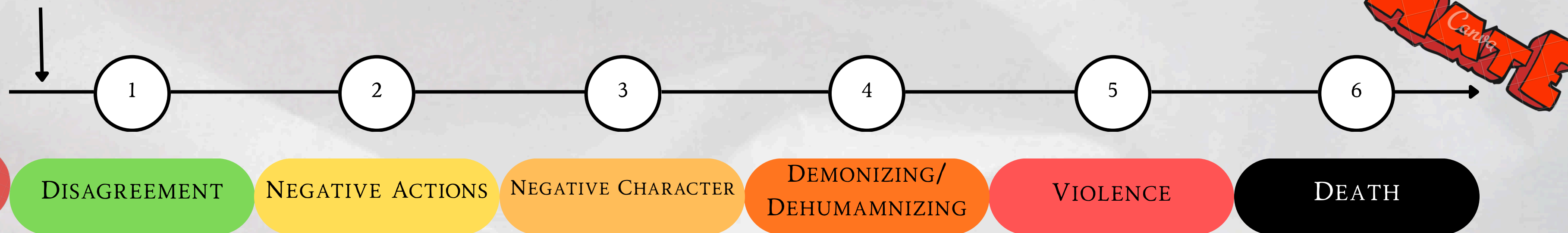




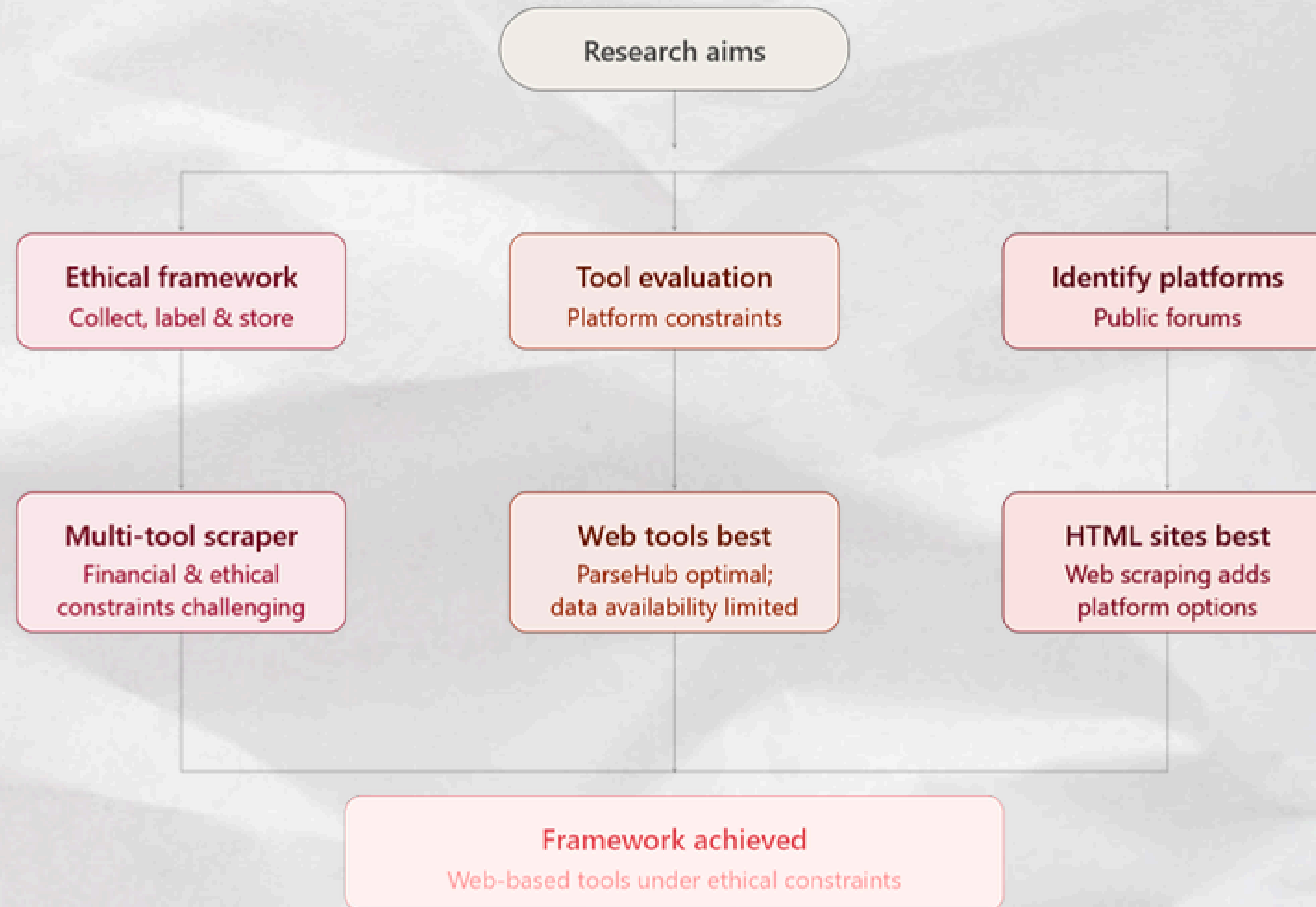
DISCUSSION OF KEY FINDINGS

- LOW HATE LEVELS: MEAN 0.28, MOSTLY NEUTRAL
- SUBTLE DEROGATION: INTELLIGENCE & CLASS HIGHEST
- NEWS SKEW: CASE, CRIME & LEGALITY DOMINATED
- PLATFORM BIAS: PER FOEDERER (2023) & YU ET AL. (2025)

MEAN 0.28



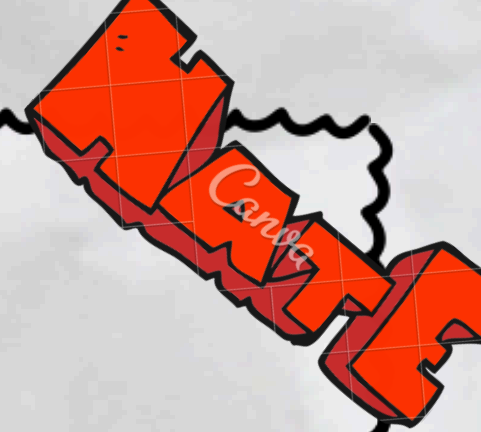
AIMS AND OBJECTIVES - OUTCOMES



○ LIMITATIONS ×

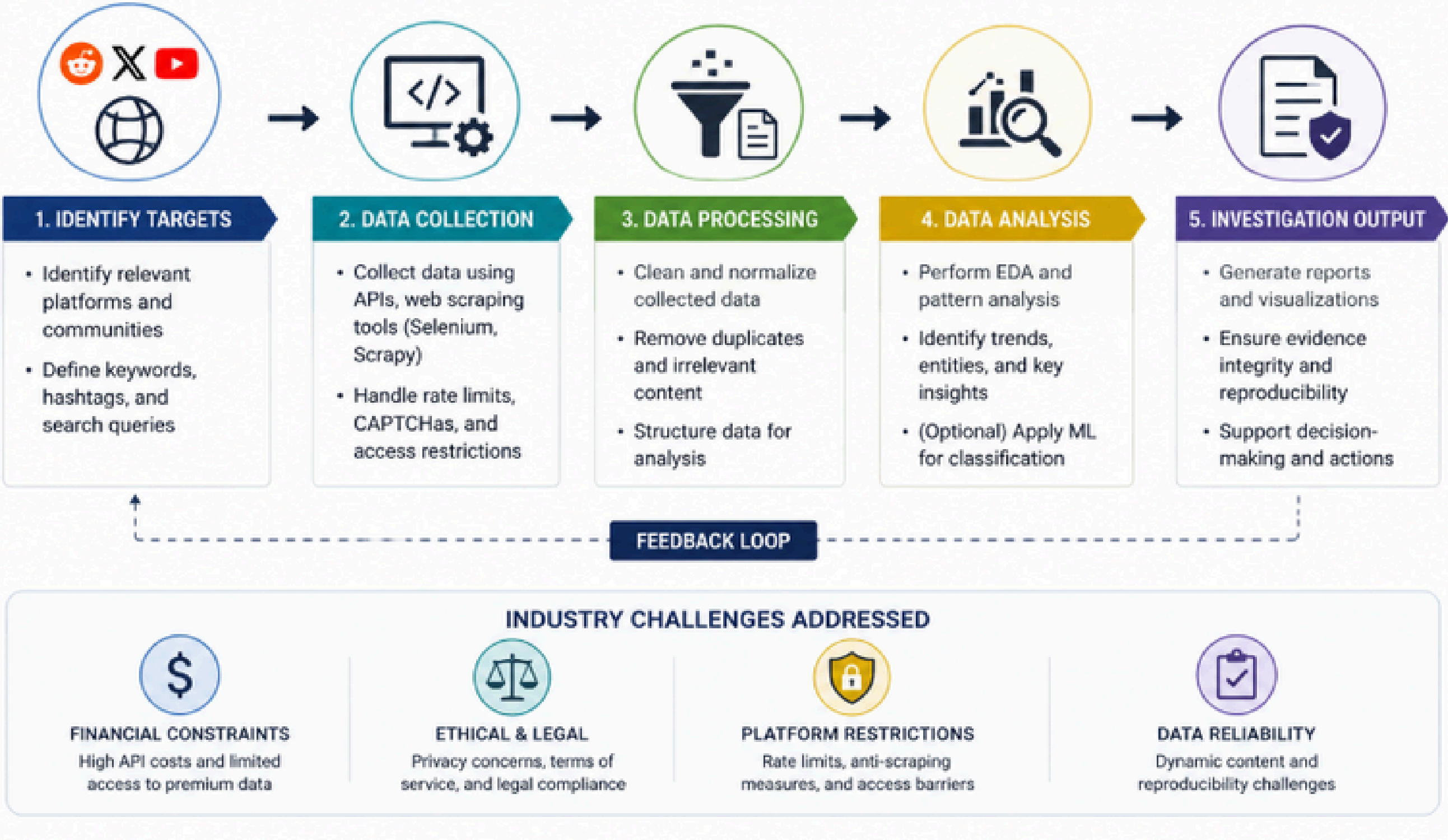
- PRACTICAL CHALLENGES OF ONLINE DATA COLLECTION
- FINANCIAL LIMITS OF APIS & COMMERCIAL TOOLS
- ETHICAL & LEGAL SCRAPING RESTRICTIONS
- TRADE-OFF: DATA ACCESS VS COMPLIANCE
- REALISTIC FRAMEWORK FOR PRACTITIONERS
- TRANSPARENT & REPRODUCIBLE INVESTIGATIONS

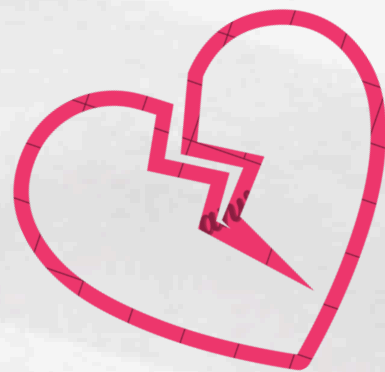




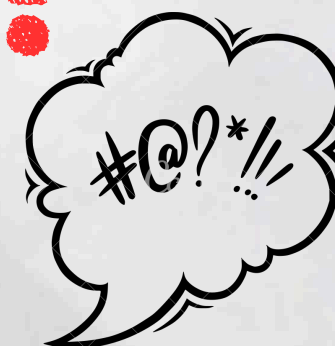
ONLINE INVESTIGATION WORKFLOW

From Data Collection to Actionable Intelligence





THX FOR WATCHING XOXO
QUESTIONS?



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